# “HEARTBEATS AND HABITS “

**24ADI204 - DATA SCIENCE AND VISUALISATION**

***Submitted by***


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***in partial fulfilment for the award of the degree of***

**BACHELOR OF TECHNOLOGY**

**IN**

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**KUMARAGURU COLLEGE OF TECHNOLOGY**

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# BONAFIDE CERTIFICATE

Certified that this project report **"Predicting 10-Year Coronary Heart Disease Risk Using the Framingham Heart Study Dataset"** is the Bonafide work of "NISHANTH P - 24BAD405, KAMALESH N -24BAD054, JAYARAKSHA REGURAJ -24BAD044, GOKULNAATH M- 24BAD026" who carried out the project work under my supervision.

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## DECLARATION

I affirm that the project work titled “HEARTBEATS AND HABITS” being submitted in partial fulfillment for the award of B. TECH Artificial Intelligence and Data Science is the original work carried out by us. It has not formed the part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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# Executive Summary

Cardiovascular disease remains one of the leading causes of mortality worldwide, with coronary heart disease (CHD) accounting for a significant proportion of deaths. Early identification of high-risk individuals enables timely interventions that can prevent or delay disease onset.

## Problem Statement

This project addresses the challenge of predicting an individual's 10-year risk of coronary heart disease using clinical, demographic, and lifestyle data. The core question is: Can a combination of biomarkers and behavioural factors accurately stratify patients into risk categories to support preventive healthcare decisions?

## Dataset Used

The Framingham Heart Study Dataset (sourced from Kaggle) was selected as it is a raw, uncleaned clinical dataset containing 4,240 patient records and 16 features, including demographic variables (age, gender), behavioural indicators (smoking, cigarettes per day), and clinical measurements (blood pressure, cholesterol, glucose, BMI). The binary target variable TenYearCHD indicates whether a patient developed CHD within 10 years.

## Key Findings

- Systolic blood pressure (sysBP) and age are the strongest and most consistent predictors of CHD risk across all analytical methods.
- Approximately 15% of the patient population (644 out of ~4,240) carry a 10-year CHD risk, indicating a clinically significant at-risk population.
- Diabetes, prevalent hypertension, and prior stroke history each dramatically amplify CHD risk — diabetic individuals show a nearly three-fold increase in risk.
- PCA confirmed substantial redundancy among blood-pressure-related features, compressing 10 features to 4 principal components while retaining ~80% of total variance.
- Tree-based AI analysis identified a glucose threshold (>121) as the single strongest nonlinear predictor, increasing CHD risk by approximately 3.13×.

## Final Insights

The analysis clearly demonstrates that CHD risk is not driven by any single factor but arises from a compounding interaction of age, blood pressure, metabolic markers, and lifestyle choices. The highest-risk cohort consists of individuals aged 60 and above who are hypertensive, diabetic, and heavy smokers. Interventions targeting blood pressure management and glucose control in the 50+ age group are most likely to yield meaningful reductions in population-level CHD incidence.

# List of Abbreviations

| Abbreviation | Full Form |
|---|---|
| CHD | Coronary Heart Disease |
| EDA | Exploratory Data Analysis |
| PCA | Principal Component Analysis |
| IQR | Interquartile Range |
| KDE | Kernel Density Estimation |
| BMI | Body Mass Index |
| sysBP | Systolic Blood Pressure |
| diaBP | Diastolic Blood Pressure |
| MI | Mutual Information |
| ML | Machine Learning |
| DSV | Data Science and Visualization |
| | |

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# Chapter 1: Introduction

## 1.1 Background of the Problem

Cardiovascular diseases (CVDs) are the leading cause of death globally, responsible for an estimated 17.9 million deaths per year according to the World Health Organization. Among CVDs, Coronary Heart Disease is particularly prevalent, arising from the narrowing or blockage of coronary arteries due to the buildup of fatty deposits. The condition often develops silently over decades, making early risk identification a clinical priority.

The Framingham Heart Study, launched in 1948 in Framingham, Massachusetts, is one of the most influential longitudinal cardiovascular studies in medical history. It produced the foundational risk factors associated with heart disease — including age, blood pressure, cholesterol, and smoking — that are still used in clinical practice today. The dataset derived from this study presents a rich real-world opportunity to apply modern data science techniques to a clinically relevant prediction problem.

## 1.2 Importance of the Study

Despite advances in cardiac medicine, many high-risk individuals go undiagnosed until they experience a serious cardiac event. A data-driven risk prediction model can supplement clinical judgment by identifying at-risk patients earlier and more systematically. This has direct public health value: targeted interventions — lifestyle modifications, medication, and monitoring — applied to correctly identified high-risk individuals can significantly reduce the burden of CHD on healthcare systems.

From a data science perspective, the Framingham dataset exemplifies real-world challenges: missing values, class imbalance, feature multicollinearity, and non-linear risk relationships. Working through these challenges end-to-end develops practical competencies in preprocessing, feature engineering, dimensionality reduction, and visual storytelling.

## 1.3 Objectives of the Project

- To perform a complete end-to-end data science workflow on the Framingham Heart Study dataset.
- To clean and preprocess raw clinical data, handling missing values, outliers, and skewed distributions.
- To conduct thorough exploratory data analysis (EDA) to uncover patterns and risk associations.
- To engineer features through scaling, encoding, and transformation for improved analytical quality.
- To identify the most predictive features using statistical and machine learning-based selection methods.
- To apply Principal Component Analysis (PCA) for dimensionality reduction and variance analysis.
- To construct an interactive Power BI dashboard that communicates insights to a non-technical audience.

- To derive actionable clinical and public health insights from the data and present them as a coherent narrative.

## 1.4 Scope of the Project

This project focuses exclusively on the analytical pipeline from raw data ingestion to insight generation and visualization. It does not include deployment of a production-grade prediction system, though the feature engineering and dimensionality reduction work lays the groundwork for one. The scope covers the full preprocessing, EDA, feature engineering, advanced analysis, and dashboard construction phases using Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn) and Power BI as the primary tools.

The dataset used is the Framingham Heart Study Dataset available on Kaggle. No additional external datasets are merged or compared. The analytical findings are interpreted within the context of existing cardiovascular medicine literature but do not constitute medical advice.

# Chapter 2: Dataset Description

## 2.1 Source of Dataset

The dataset used in this project is the Framingham Heart Study Dataset, sourced from Kaggle at the following URL:
https://www.kaggle.com/datasets/dileep070/heart-disease-prediction-using-logistic-regression
This dataset is derived from the landmark Framingham Heart Study, a long-running cardiovascular cohort study. It was selected from four candidate datasets after a structured evaluation process, as it best satisfies the project requirements: it is raw and uncleaned (enabling a full preprocessing demonstration), contains a sufficient number of attributes and records for meaningful analysis, and is well-aligned with the project goal of CHD risk prediction.

```
import numpy as np
import pandas as pd
df=pd.read_csv("./data/framingham.csv")
```

```
df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4238 entries, 0 to 4237
Data columns (total 16 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   male             4238 non-null   int64
 1   age              4238 non-null   int64
 2   education        4133 non-null   float64
 3   currentSmoker    4238 non-null   int64
 4   cigsPerDay       4209 non-null   float64
 5   BPMeds           4185 non-null   float64
 6   prevalentStroke  4238 non-null   int64
 7   prevalentHyp     4238 non-null   int64
 8   diabetes         4238 non-null   int64
 9   totChol          4188 non-null   float64
 10  sysBP            4238 non-null   float64
 11  diaBP            4238 non-null   float64
 12  BMI              4219 non-null   float64
 13  heartRate        4237 non-null   float64
 14  glucose          3850 non-null   float64
 15  TenYearCHD       4238 non-null   int64
dtypes: float64(9), int64(7)
memory usage: 529.9 KB
```

## 2.2 Dataset Dimensions

| Property | Value |
|---|---|
| Total Records (Rows) | 4,240 |
| Total Features (Columns) | 16 |
| Target Variable | TenYearCHD (Binary: 0 = No CHD, 1 = CHD) |
| Positive Class (CHD = 1) | ~644 records (~15.2%) |
| Negative Class (CHD = 0) | ~3,596 records (~84.8%) |
| Data Format | CSV (.csv) |

## 2.3 Feature Description

| Feature | Type | Description |
|---|---|---|
| male | Categorical (Binary) | Gender (1 = Male, 0 = Female) |
| age | Continuous | Age of patient in years |
| education | Ordinal | Education level (1–4 scale: Some HS to College) |
| currentSmoker | Categorical (Binary) | Whether the patient is a current smoker |
| cigsPerDay | Continuous | Average number of cigarettes smoked per day |
| BPMeds | Categorical (Binary) | Whether on blood pressure medication |
| prevalentStroke | Categorical (Binary) | History of stroke (1 = Yes) |
| prevalentHyp | Categorical (Binary) | Prevalent hypertension (1 = Yes) |
| diabetes | Categorical (Binary) | Diabetic status (1 = Yes) |
| totChol | Continuous | Total cholesterol level (mg/dL) |
| sysBP | Continuous | Systolic blood pressure (mmHg) |
| diaBP | Continuous | Diastolic blood pressure (mmHg) |
| BMI | Continuous | Body Mass Index (kg/m²) |
| heartRate | Continuous | Resting heart rate (beats per minute) |
| glucose | Continuous | Fasting blood glucose level (mg/dL) |
| TenYearCHD | Target (Binary) | 10-year CHD risk (1 = High Risk, 0 = Low Risk) |

## 2.4 Initial Data Issues

Upon initial inspection, the following data quality concerns were identified:

- Missing Values: Seven columns contain null values — education, cigsPerDay, BPMeds, totChol, BMI, heartRate, and glucose. These require imputation before analysis.
- Class Imbalance: The target variable TenYearCHD is imbalanced, with approximately 85% negative cases and 15% positive cases. This may affect model performance and requires consideration during evaluation.
- Feature Scale Variation: Feature values vary widely in scale (e.g., age ranges 32–70, while glucose ranges 40–394), necessitating normalization before any distance-based analysis.
- Outliers: Several continuous features — particularly glucose, sysBP, diaBP, and BMI — contain extreme values that could distort statistical summaries and model fitting.
- Skewed Distributions: Features such as cigsPerDay and sysBP exhibit significant right-skewness, violating normality assumptions required by linear models.
- No Duplicate Rows: Initial inspection confirmed no duplicate records in the dataset.

# Chapter 3: Data Preprocessing and Cleaning

## 3.1 Data Loading and Inspection

The dataset was loaded into a Pandas DataFrame using pd.read_csv(). Initial inspection involved examining the shape (4240 rows × 16 columns), checking data types using df.info(), and computing basic statistics using df.describe(). The target variable TenYearCHD was confirmed as a binary integer column. All other features were stored as int64 or float64, with no categorical string variables requiring encoding at this stage.

## 3.2 Handling Missing Values

Missing values were identified and addressed using a two-step imputation strategy based on the data type of each affected column.

```
df.isnull().sum()
```

```
male                 0
age                  0
education          105
currentSmoker        0
cigsPerDay          29
BPMeds              53
prevalentStroke      0
prevalentHyp         0
diabetes             0
totChol             50
sysBP                0
diaBP                0
BMI                 19
heartRate            1
glucose            388
TenYearCHD           0
dtype: int64
```

| Column | Missing Count | Data Type | Imputation Method | Justification |
|---|---|---|---|---|
| education | 105 | Ordinal | Mode | Ordinal categorical — mode preserves the most common category |
| cigsPerDay | 29 | Continuous | Median | Right-skewed; median is robust to extreme smoker values |
| BPMeds | 53 | Binary | Mode | Binary flag — mode replaces with the dominant class |
| totChol | 50 | Continuous | Median | Medical values may have outliers; median is appropriate |

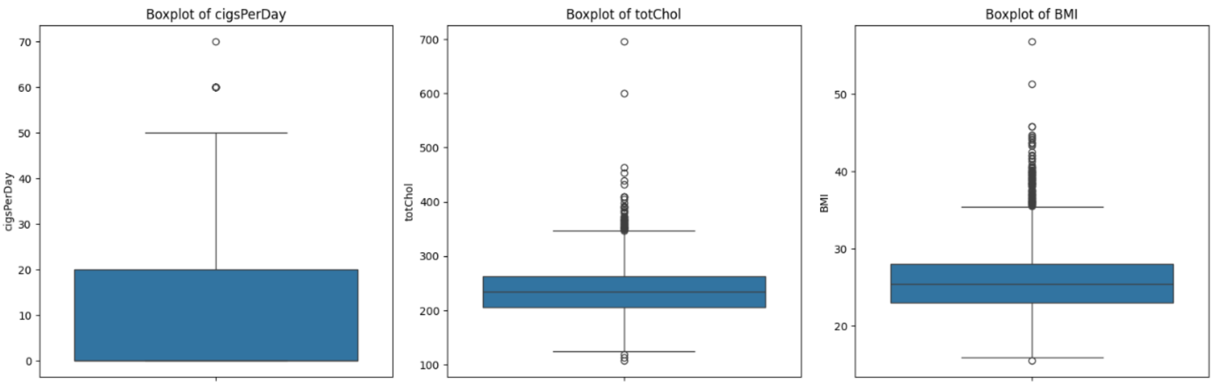
| Column | Missing Count | Data Type | Imputation Method | Justification |
|---|---|---|---|---|
| BMI | 19 | Continuous | Median | Slight skewness and outliers make median safer than mean |
| heartRate | 1 | Continuous | Median | Single missing value; median imputation is safe here |
| glucose | 388 | Continuous | Median | Largest missing count; high right-skew makes median critical |

Median was chosen over mean for all continuous features because medical data frequently contains extreme values. The median, being resistant to outliers, prevents imputed values from being dragged toward unrealistic extremes.

## 3.3 Outlier Detection

Outlier detection was performed using two complementary approaches:

```
plt.figure(figsize=(16, 20))

for i, col in enumerate(numerical_cols, 1):
    plt.subplot(4, 3, i)
    sns.boxplot(y=df[col])
    plt.title(f"Boxplot of {col}")

plt.tight_layout()
plt.show()
```




### Z-Score Method

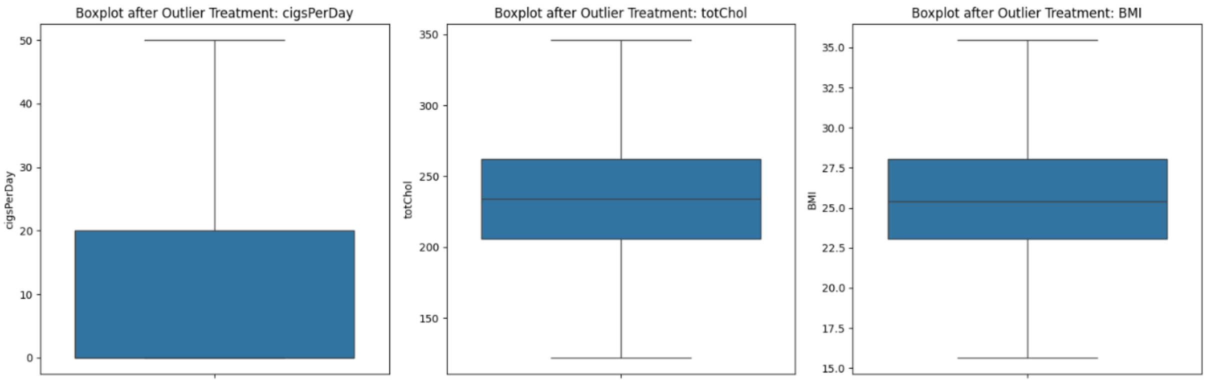
Z-scores were calculated for all continuous features. Values with $|Z| > 3$ were flagged as outliers. Features with the highest outlier counts included BMI, glucose, sysBP, and diaBP, confirming their heavy-tailed distributions.

### IQR-Based Capping

Boxplots visually confirmed the outlier-heavy nature of glucose, diaBP, sysBP, and BMI. Rather than removing these records (which would reduce the already limited positive-class samples), IQR-based capping was applied. For each affected column, values below Q1 – 1.5×IQR were clipped to the lower bound, and values above Q3 + 1.5×IQR were clipped to the upper bound. This preserves the record while constraining the impact of extreme values.

- Verify Outliers After Treatment

```
plt.figure(figsize=(16, 20))

for i, col in enumerate(outlier_cols, 1):
    plt.subplot(4, 3, i)
    sns.boxplot(y=df[col])
    plt.title(f"Boxplot after Outlier Treatment: {col}")

plt.tight_layout()
plt.show()
```




## 3.4 Data Quality Report Summary

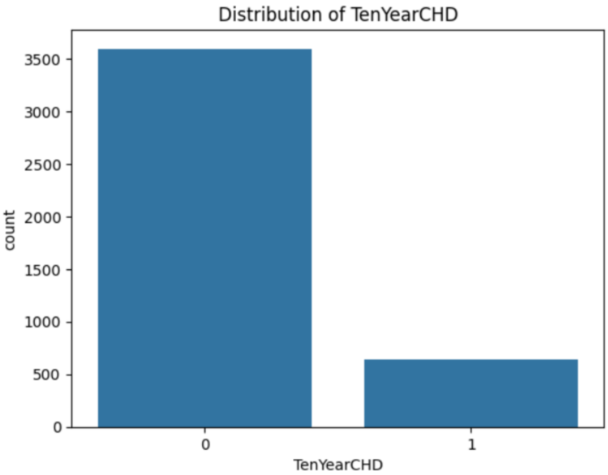
| Metric | Before Cleaning | After Cleaning |
|---|---|---|
| Total Missing Values | 645 | 0 |
| Duplicate Rows | 0 | 0 |
| Outlier Flags (Z > 3) | Present in 4 features | Capped via IQR |
| Columns with Null Values | 7 | 0 |
| Dataset Shape | 4240 × 16 | 4240 × 16 |

# Chapter 4: Exploratory Data Analysis (EDA)

## 4.1 Univariate Analysis

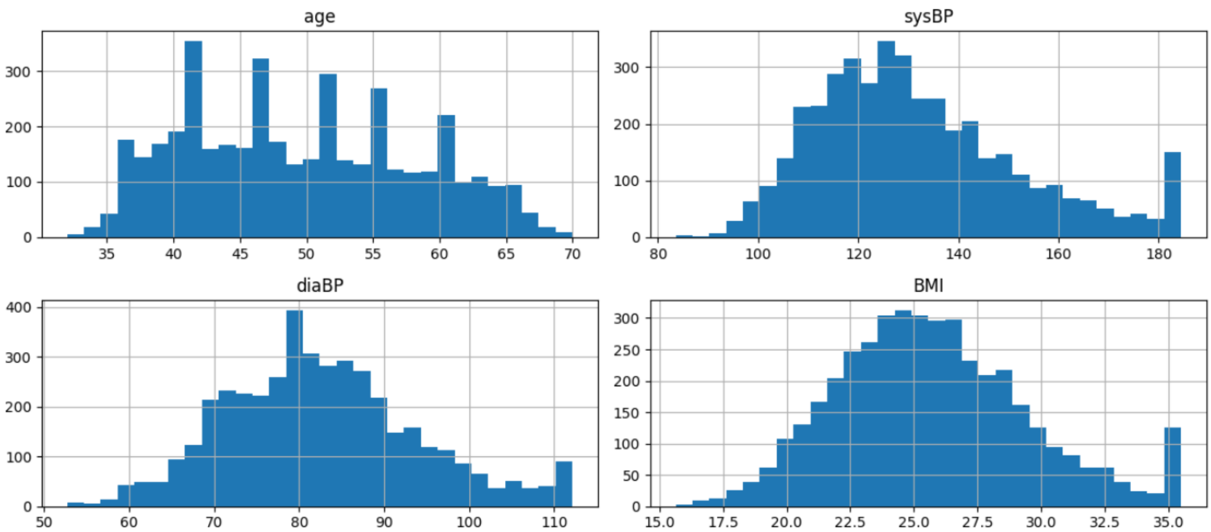
Univariate analysis examines the distribution of each variable independently. Histograms were plotted for all continuous variables.

```
sns.countplot(x='TenYearCHD', data=df)
plt.title("Distribution of TenYearCHD")
plt.show()
```




**Continuous Variable Distribution**

```
num_cols = ['age', 'sysBP', 'diaBP', 'BMI', 'glucose', 'totChol']

df[num_cols].hist(bins=30, figsize=(12,8))
plt.tight_layout()
plt.show()
```



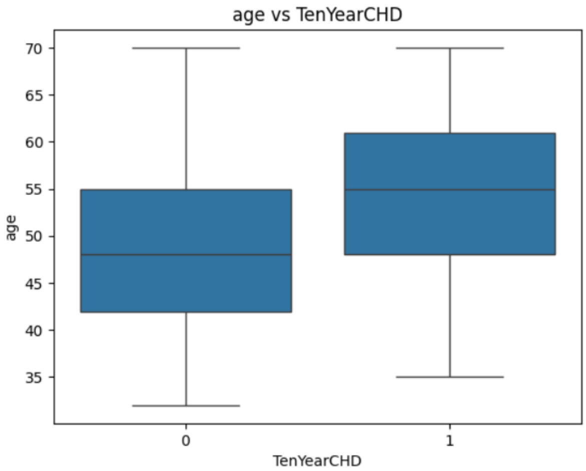

The target variable TenYearCHD shows a pronounced class imbalance: approximately 3,596 patients (84.8%) are negative for CHD and 644 (15.2%) are positive. Most continuous features exhibit right-skewed distributions. cigsPerDay shows the strongest right skew, attributable to the high proportion of non-smokers (zero cigarettes) alongside a subset of heavy smokers. sysBP and glucose also display significant right tails indicating the presence of hypertensive and hyperglycaemic outliers respectively. Age, diaBP, BMI, and totChol show near-normal bell-shaped distributions with mild right skew.

## 4.2 Bivariate Analysis

### Numerical Features vs Target

**Numerical Feature vs Target**

```
for col in ['age', 'sysBP', 'BMI', 'glucose']:
    sns.boxplot(x='TenYearCHD', y=col, data=df)
    plt.title(f"{col} vs TenYearCHD")
    plt.show()
```

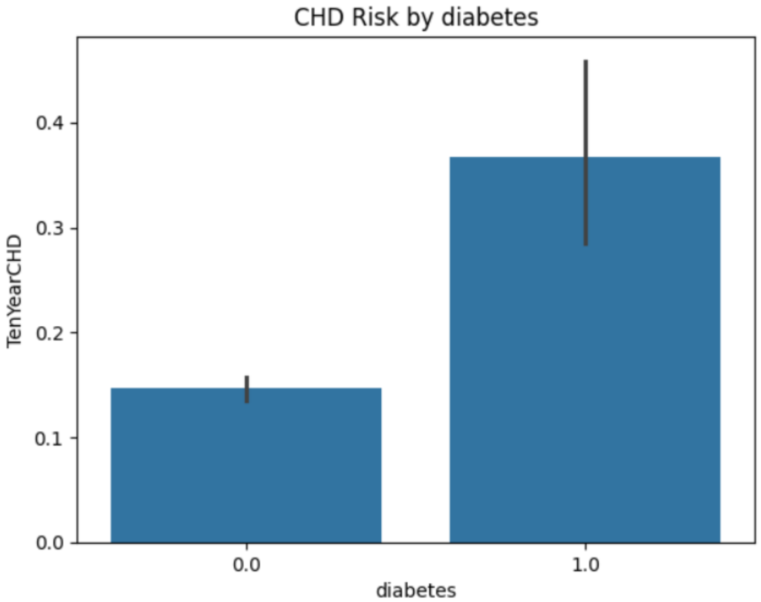



Box plots comparing each continuous feature against TenYearCHD revealed the following key relationships:

- Age: CHD patients have a noticeably higher median age (~55) compared to non-CHD patients (~48). The entire age distribution for CHD cases is shifted rightward, confirming age as a strong positive predictor.
- Systolic Blood Pressure (sysBP): The CHD group shows a higher median sysBP (~140 vs ~125) and a wider spread, with extreme values reaching above 300 mmHg. This is among the strongest distinguishing features.
- BMI: CHD and non-CHD groups show only marginally different median BMI values (~26 vs ~25), with heavy overlap in their interquartile ranges. BMI is a weak individual predictor in this dataset.
- Glucose: Distributions for both groups nearly overlap (median ~77–78), indicating glucose has limited standalone discriminative power, though extreme values may carry nonlinear signal.

### Categorical Features vs Target

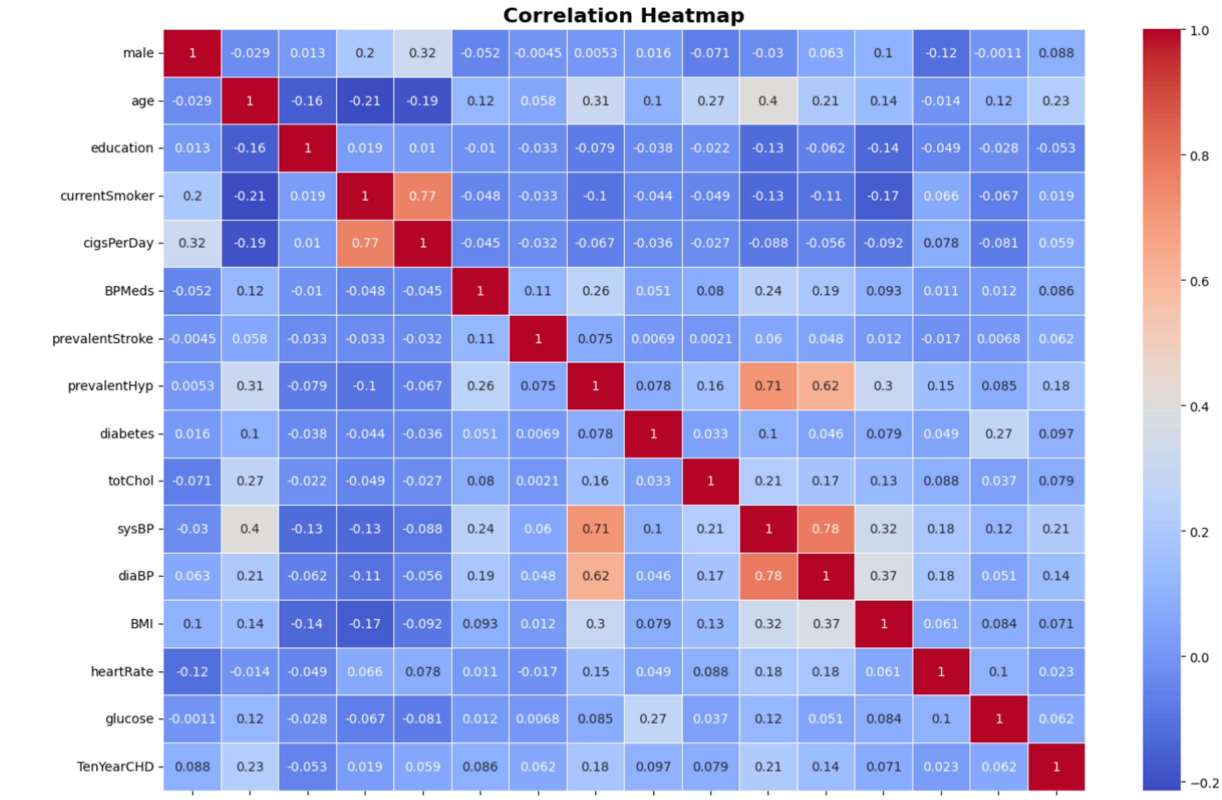
```
cat_cols = ['male', 'currentSmoker', 'diabetes', 'prevalentHyp']

for col in cat_cols:
    sns.barplot(x=col, y='TenYearCHD', data=df)
    plt.title(f"CHD Risk by {col}")
    plt.show()
```




Bar charts of average CHD rate across categorical features revealed:

- Gender (male): Males exhibit a CHD rate of approximately 19%, compared to ~12–13% in females — a clinically meaningful difference.
- Diabetes: Diabetic individuals show a dramatically elevated CHD rate (~37%) versus non-diabetics (~14–15%), making diabetes one of the most discriminating binary features.
- Prevalent Hypertension: Hypertensive patients show ~25% CHD rate vs ~11% in non-hypertensives — a strong categorical predictor.
- Current Smoker: Smokers show slightly elevated CHD rate (~16% vs ~14.5%). The gap is modest, with overlapping confidence intervals, suggesting smoking interacts with other variables rather than acting as a standalone driver.

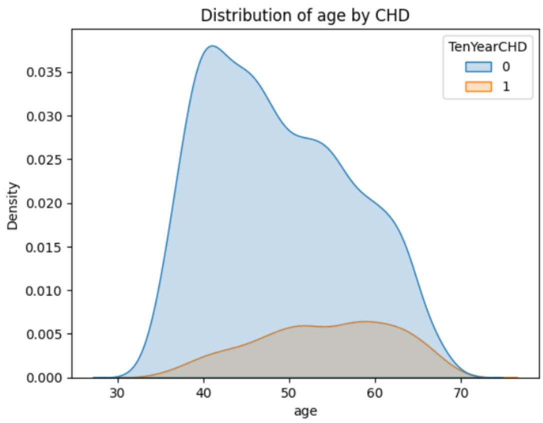
### 4.3 Multivariate Analysis




A correlation heatmap revealed strong multicollinearity among blood pressure features: sysBP and diaBP correlate at approximately 0.78, and both strongly correlate with prevalentHyp. The currentSmoker and cigsPerDay features are nearly redundant, with a correlation of ~0.77. Against the target TenYearCHD, the strongest linear correlations are age (~0.23), sysBP (~0.22), and prevalentHyp (~0.18), confirming that no single feature dominates and prediction inherently requires multivariate modelling.

**KDE Distribution by Target**

```
for col in ['age', 'sysBP', 'glucose']:
    sns.kdeplot(data=df, x=col, hue='TenYearCHD', fill=True)
    plt.title(f"Distribution of {col} by CHD")
    plt.show()
```




KDE plots by target class visually reinforced that age and sysBP distributions are meaningfully shifted between CHD and non-CHD groups, while glucose distributions nearly completely overlap — consistent with the bivariate findings.

# Chapter 5: Data Transformation and Feature Engineering

## 5.1 Skewness Detection and Correction

```
skewness = df[continuous_cols].skew().sort_values(ascending=False)
print("Skewness of continuous features:\n")
print(skewness.to_string())
```

```
Skewness of continuous features:

cigsPerDay    1.180184
sysBP         0.732407
glucose       0.449353
BMI           0.438295
diaBP         0.404980
heartRate     0.403390
totChol       0.330328
age           0.228146
```

Skewness was computed for all continuous features. cigsPerDay exhibited the highest skewness (>1.18), followed by sysBP (~0.7). Features with skewness above 0.5 were targeted for transformation.

The log1p transformation — defined as log(1 + x) — was applied to cigsPerDay. This function was selected over pure log() because it handles zero values safely (critical for non-smokers with cigsPerDay = 0), avoids undefined outputs, and effectively compresses the right tail. After transformation, the skewness of cigsPerDay reduced from 1.18 to approximately 0.32, making it substantially more suitable for linear models.

## 5.2 Categorical Encoding

```
print("Before:", df['education'].dtype, "| Sample:", df['education'].unique())

df['education'] = df['education'].astype(int)

print("After :", df['education'].dtype, "| Sample:", sorted(df['education'].unique()))
```

```
Before: float64 | Sample: [4. 2. 1. 3.]
After : int64 | Sample: [np.int64(1), np.int64(2), np.int64(3), np.int64(4)]
```

The education feature uses an ordinal scale (1 = Some High School, 2 = High School Graduate, 3 = Some College, 4 = College Graduate). Ordinal Encoding was applied rather than One-Hot Encoding because the categories carry a meaningful natural ordering — the model should learn that education level 4 is greater than level 1, not treat them as independent binary attributes. The encoded distribution confirmed that lower education levels (1 and 2) are disproportionately represented, indicating a dataset skew toward less-educated individuals.

### 5.3 Feature Scaling

```
scale_cols = ['age', 'totChol', 'sysBP', 'diaBP', 'BMI', 'heartRate', 'glucose', 'cigsPerDay']

scaler = RobustScaler()
df[scale_cols] = scaler.fit_transform(df[scale_cols])

print("After RobustScaler — descriptive statistics:")
df[scale_cols].describe().round(3)
```

```
After RobustScaler — descriptive statistics:
```

| | age | totChol | sysBP | diaBP | BMI | heartRate | glucose | cigsPerDay |
|---|---|---|---|---|---|---|---|---|
| count | 4238.000 | 4238.000 | 4238.000 | 4238.000 | 4238.000 | 4238.000 | 4238.000 | 4238.000 |
| mean | 0.042 | 0.039 | 0.145 | 0.049 | 0.064 | 0.050 | 0.112 | 0.441 |
| std | 0.612 | 0.755 | 0.765 | 0.766 | 0.764 | 0.775 | 0.881 | 0.482 |
| min | -1.214 | -2.000 | -1.648 | -1.971 | -1.968 | -1.967 | -1.962 | 0.000 |
| 25% | -0.500 | -0.500 | -0.407 | -0.471 | -0.468 | -0.467 | -0.462 | 0.000 |
| 50% | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 75% | 0.500 | 0.500 | 0.593 | 0.529 | 0.532 | 0.533 | 0.538 | 1.000 |
| max | 1.500 | 2.000 | 2.093 | 2.029 | 2.032 | 2.033 | 2.038 | 1.291 |

Three scaling methods were considered: StandardScaler (mean/std — sensitive to outliers), MinMaxScaler (min/max — highly sensitive to extremes), and RobustScaler (median/IQR — resistant to outliers). Given that the dataset contains confirmed extreme values in glucose, sysBP, diaBP, and BMI, RobustScaler was selected as the most appropriate method.

| Scaler | Formula | Outlier Sensitivity | Selected? |
|---|---|---|---|
| StandardScaler | (x − mean) / std | High | No |
| MinMaxScaler | (x − min) / (max − min) | Very High | No |
| RobustScaler | (x − median) / IQR | Low | Yes ✓ |

After scaling, all features were centered near zero with comparable ranges. Crucially, the relative ordering and distribution shapes were preserved — RobustScaler does not alter skewness, so the prior log1p transformation on cigsPerDay remained important.

### 5.4 Before vs After Comparison

| Feature | Skewness (Before) | Skewness (After log1p + scaling) |
|---|---|---|
| cigsPerDay | 1.18 | ~0.32 |
| sysBP | ~0.70 | ~0.70 (scaling preserves shape) |
| glucose | ~0.45 | ~0.45 |
| BMI | ~0.43 | ~0.43 |

# Chapter 6: Advanced Analysis

## 6.1 Correlation Analysis

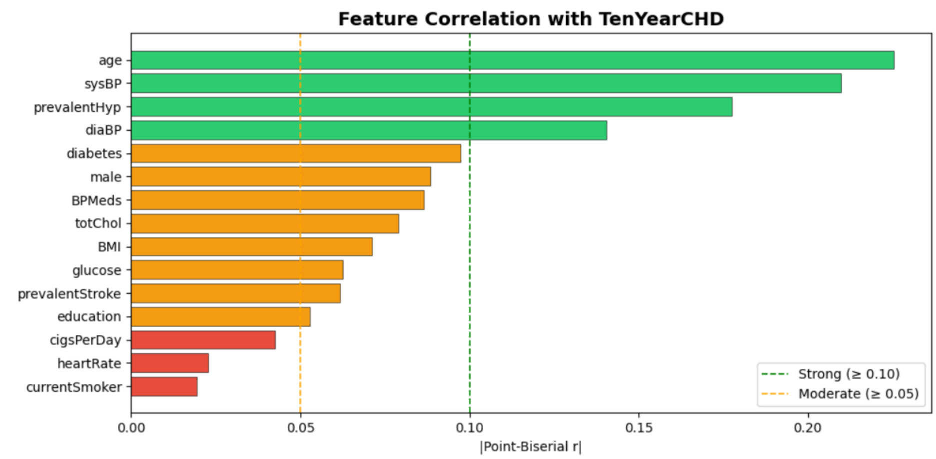
A full correlation heatmap was computed on the preprocessed feature set. Key observations include: strong multicollinearity between sysBP, diaBP, and prevalentHyp (correlations 0.62–0.78), and moderate positive linear correlations between age and CHD (~0.23) and sysBP and CHD (~0.22). No feature exceeded a 0.25 correlation with the target, confirming CHD as an inherently multivariate problem.

## 6.2 Feature Selection

Three complementary feature selection methods were applied and their rankings compared:

```
corr_scores = {}
for col in feature_cols:
    r, p = pointbiserialr(y, X[col])
    corr_scores[col] = abs(r)   # use absolute value — direction doesn't matter for ranking

corr_df = pd.DataFrame({"feature": list(corr_scores.keys()),
                        "corr": list(corr_scores.values())}).sort_values("corr", ascending=False)
print(corr_df.to_string(index=False))
```

```
        feature     corr
            age 0.225256
          sysBP 0.209686
   prevalentHyp 0.177603
          diaBP 0.140473
       diabetes 0.097317
           male 0.088428
         BPMeds 0.086417
        totChol 0.079056
            BMI 0.071243
        glucose 0.062400
prevalentStroke 0.061810
      education 0.052812
     cigsPerDay 0.042602
      heartRate 0.022828
  currentSmoker 0.019456
```



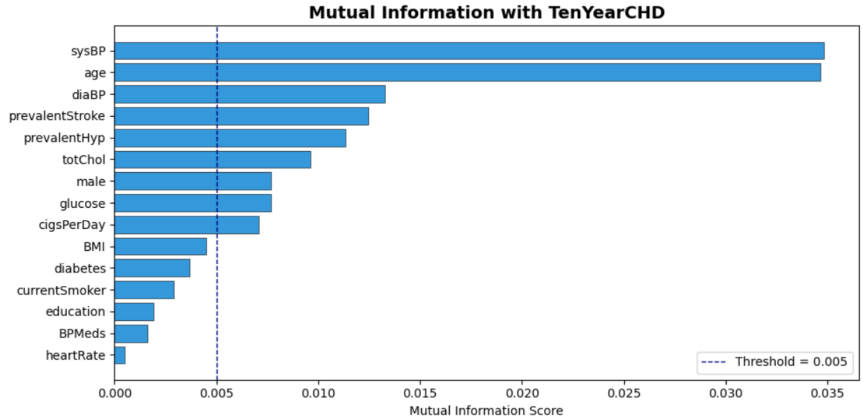

### Point-Biserial Correlation (Linear)

Measures linear association between each continuous feature and the binary target. Top predictors: age, sysBP, prevalentHyp, diaBP. Weak predictors: cigsPerDay, heartRate, currentSmoker. A threshold of 0.10 was used to classify strong vs moderate predictors.

## Mutual Information (Nonlinear)

```
mi_scores = mutual_info_classif(X, y, random_state=42)
mi_df = pd.DataFrame({"feature": feature_cols, "MI": mi_scores}).sort_values("MI", ascending=False)
print(mi_df.to_string(index=False))
```

```
       feature       MI
         sysBP 0.034811
           age 0.034659
         diaBP 0.013259
prevalentStroke 0.012476
  prevalentHyp 0.011348
       totChol 0.009608
          male 0.007684
       glucose 0.007662
    cigsPerDay 0.007074
           BMI 0.004518
      diabetes 0.003688
 currentSmoker 0.002898
     education 0.001929
        BPMeds 0.001620
     heartRate 0.000509
```

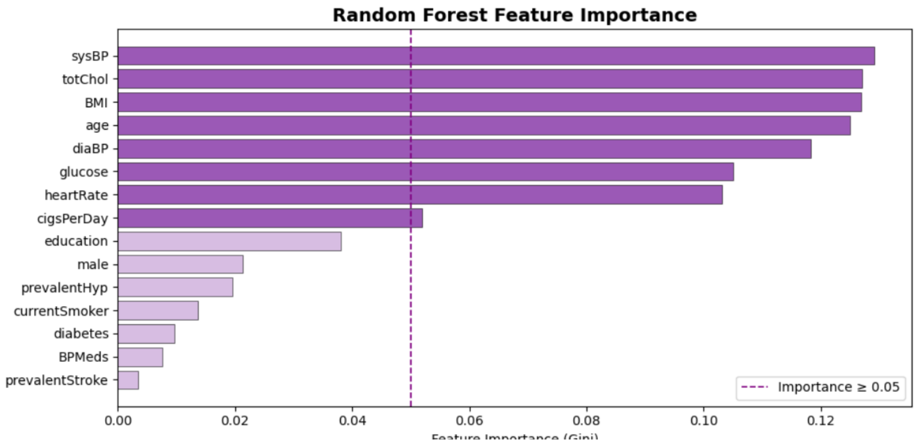



Captures nonlinear dependencies. sysBP and age remained top-ranked, confirming their importance beyond linear effects. prevalentStroke appeared more important here than in correlation analysis, indicating a nonlinear or interaction effect. heartRate, BPMeds, and education showed near-zero MI.

## Random Forest Feature Importance (Tree-Based)

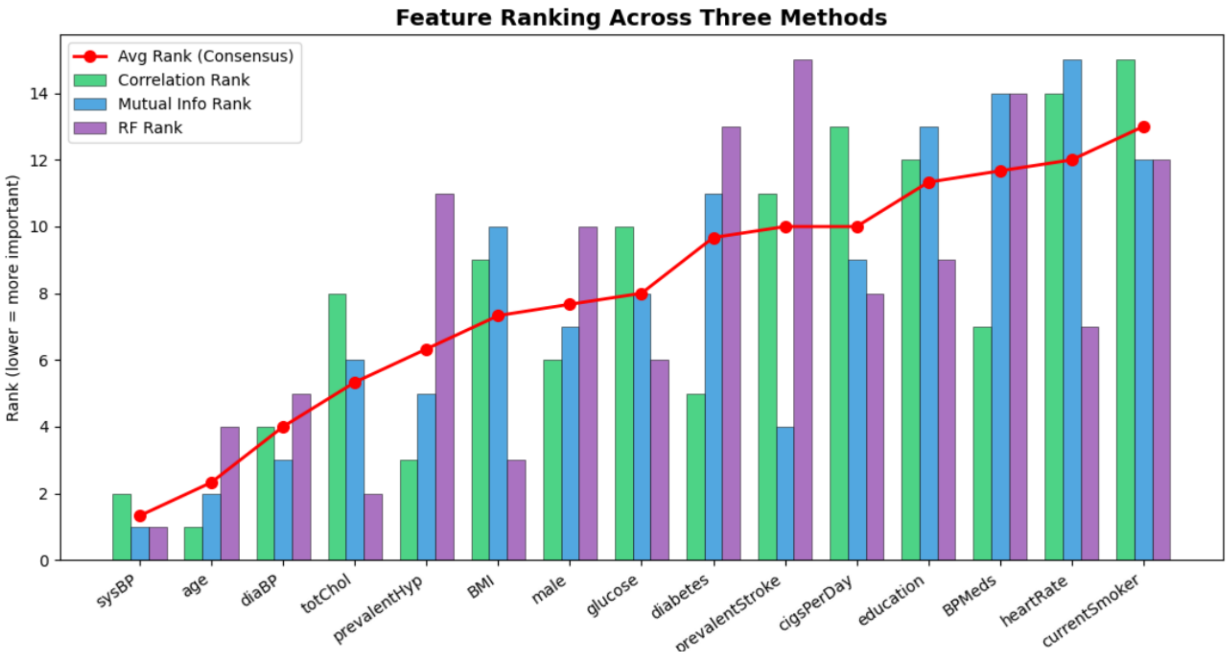
```
rf = RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)
rf.fit(X, y)

rf_df = pd.DataFrame({"feature": feature_cols,
                      "importance": rf.feature_importances_}).sort_values("importance", ascending=False)
print(rf_df.to_string(index=False))
```

```
       feature  importance
         sysBP    0.129061
       totChol    0.127067
           BMI    0.126883
           age    0.125064
         diaBP    0.118283
       glucose    0.105078
     heartRate    0.103098
    cigsPerDay    0.051976
     education    0.038001
          male    0.021313
  prevalentHyp    0.019595
 currentSmoker    0.013721
      diabetes    0.009724
        BPMeds    0.007635
prevalentStroke    0.003499
```




Captures both linear and interaction effects. BMI, glucose, and heartRate gained relative importance compared to correlation-based methods, suggesting they contribute through nonlinear interactions. The importance was distributed more evenly, reflecting the ensemble's ability to exploit weak signals.

### Consensus Ranking and Final Feature Selection




| Feature | Correlation Rank | MI Rank | RF Rank | Consensus Decision |
|---|---|---|---|---|
| sysBP | 1 | 1 | 1 | Retain — top predictor across all methods |
| age | 2 | 2 | 4 | Retain — consistently high |
| diaBP | 4 | 3 | 5 | Retain — strong across methods |
| totChol | 6 | 5 | 2 | Retain — important in RF |
| prevalentHyp | 3 | 4 | 8 | Retain — clinically important |

| Feature | Correlation Rank | MI Rank | RF Rank | Consensus Decision |
|---|---|---|---|---|
| BMI | 7 | 7 | 3 | Retain — nonlinear signal |
| glucose | 8 | 8 | 6 | Retain — nonlinear importance |
| male | 5 | 6 | 9 | Retain — demographic predictor |
| diabetes | 9 | 9 | 11 | Retain — clinical significance |
| cigsPerDay | 10 | 10 | 7 | Retain — moderate nonlinear signal |
| currentSmoker | 11 | 11 | 10 | Drop — redundant with cigsPerDay |
| heartRate | 13 | 12 | 6 | Drop — consistently weak |
| education | 15 | 15 | 13 | Drop — minimal contribution |
| BPMeds | 12 | 14 | 12 | Drop — weak signal |
| prevalentStroke | 14 | 13 | 14 | Drop — rare and weak standalone |

The final selected feature set (10 features) reduces dimensionality from 15 to 10 while retaining the highest-signal variables across all three methods.

### 6.3 Principal Component Analysis (PCA)

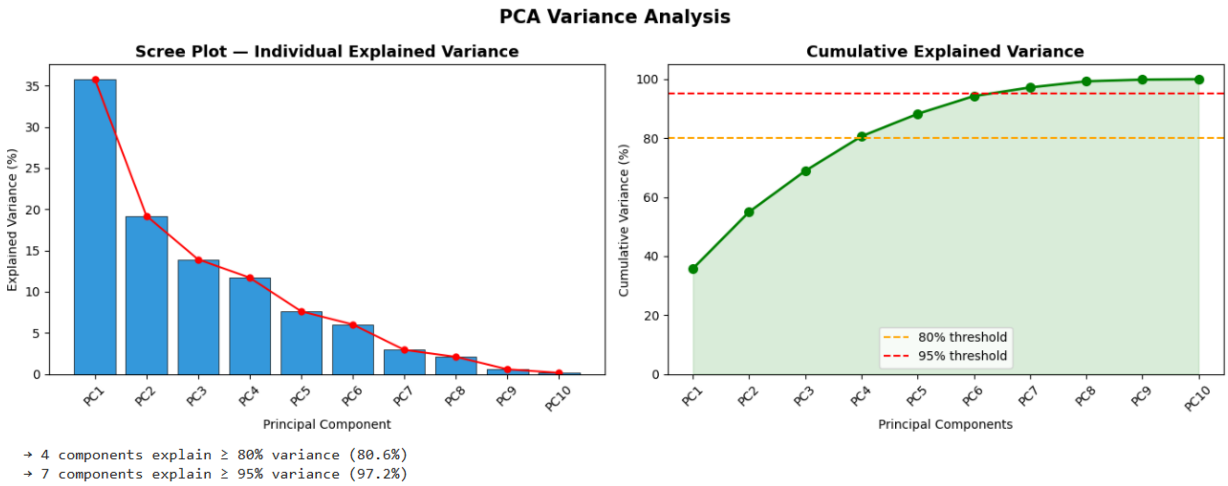
PCA was applied to the 10 selected, scaled features to explore the underlying variance structure of the dataset.

```
pca = PCA(random_state=42)
pca.fit(X_selected)

explained_var  = pca.explained_variance_ratio_
cumulative_var = np.cumsum(explained_var)

pca_summary = pd.DataFrame({
    "Component"      : [f"PC{i+1}" for i in range(len(explained_var))],
    "Explained Var %" : (explained_var * 100).round(2),
    "Cumulative %"    : (cumulative_var * 100).round(2)
})
print(pca_summary.to_string(index=False))
```

**Explained Variance**

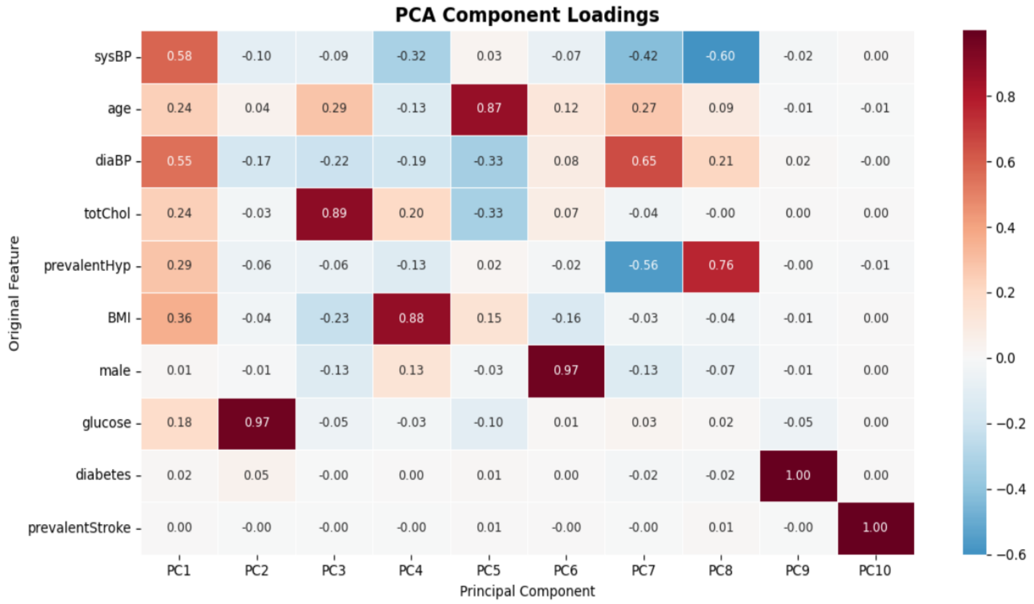
| Principal Component | Individual Variance Explained | Cumulative Variance |
|---|---|---|
| PC1 | ~36% | ~36% |
| PC2 | ~18% | ~54% |

| Principal Component | Individual Variance Explained | Cumulative Variance |
|---|---|---|
| PC3 | ~15% | ~69% |
| PC4 | ~11% | ~80% |
| PC5 | ~8% | ~88% |
| PC6 | ~6% | ~94% |
| PC7 | ~3% | ~97% |
| PC8 | ~2% | ~99% |
| PC9 | ~1% | ~100% |




The scree plot shows a clear elbow at PC3–PC4, indicating that 4 principal components capture ~80% of total variance — a practical dimensionality reduction from 10 to 4 features.

### Component Interpretation (Loading Heatmap)




```
Interpretation:
  PC1 (35.8% var): mainly driven by ['sysBP', 'diaBP', 'BMI']
  PC2 (19.2% var): mainly driven by ['glucose', 'diaBP', 'sysBP']
  PC3 (13.9% var): mainly driven by ['totChol', 'age', 'BMI']
```

- PC1 – Cardiovascular Pressure Factor: High loadings on sysBP (~0.58) and diaBP (~0.55). Represents overall blood pressure and body condition.
- PC2 – Metabolic Factor: Dominated by glucose (~0.97). Captures the diabetic/metabolic signal independently.
- PC3 – Lipid and Aging Factor: Strong loading on totChol (~0.89) with contribution from age. Represents lipid profile and aging risk.

The 2D scatter and 3D scatter of PCA components show heavy overlap between CHD and non-CHD classes, confirming that PCA — being an unsupervised variance-maximising method — does not separate classes well. This reinforces the need for supervised learning models (Logistic Regression, Random Forest, XGBoost) for actual classification.

**Engineering Judgment**

Given that feature selection already reduced the dimensionality to 10 interpretable clinical features, and that PCA reduces interpretability (original feature names are lost in principal components), retaining the original selected features is recommended for any clinical or explainable AI application. PCA is more beneficial in high-dimensional datasets (100+ features); here, its primary value was confirming feature redundancy rather than as a preprocessing step for the final model.

# Chapter 7: Visualization and Dashboard

## 7.1 Tools Used

The following tools were employed across the project's visualization pipeline:

| Tool / Library | Purpose |
|---|---|
| Python – Matplotlib | Static plots: histograms, bar charts, line plots |
| Python – Seaborn | Statistical visualizations: box plots, KDE, heatmaps, pair plots |
| Python – Scikit-learn | PCA computation and feature importance (Random Forest) |
| Power BI | Interactive dashboard construction and KPI reporting |

## 7.2 Types of Visualizations Used

| Visualization Type | Purpose | Chapter Used |
|---|---|---|
| Histogram | Show univariate distribution shapes and skewness | Chapter 4 |
| Box Plot | Compare feature distributions across CHD target classes | Chapter 4 |
| KDE Plot | Smooth density estimation by target class | Chapter 4 |
| Bar Chart | Categorical feature vs CHD risk rates | Chapter 4 |
| Correlation Heatmap | Visualize pairwise feature correlations and multicollinearity | Chapter 4, 6 |
| Pair Plot | Multivariate pairwise scatter with class colouring | Chapter 4 |
| Scree Plot | Explained variance per PCA component | Chapter 6 |
| PCA Loading Heatmap | Component loadings to interpret PCA dimensions | Chapter 6 |
| 2D / 3D Scatter | PCA component visualizations with class labels | Chapter 6 |
| Power BI Dashboard | Interactive KPI cards, trend lines, risk segments | Chapter 7 |

## 7.3 Dashboard Explanation

The Power BI dashboard was structured into four panels to tell the complete CHD risk story from population overview to individual risk pathways:

**Panel 1 – KPI Overview**

Displays four summary KPI cards: total patient count (~4,240), overall 10-year CHD risk rate (~15.2%), average patient age (~50 years), and average cholesterol (~236 mg/dL). These anchor the audience with population-level context before drilling into risk factors.

**Panel 2 – Demographic and Lifestyle Risk Factors**

Includes age-stratified CHD risk bar charts, gender comparison (rate-normalized), education-level trend, smoking status breakdown, and a dose-response line chart of cigsPerDay vs CHD risk. The dose-response chart is particularly informative: risk increases monotonically with cigarette intensity, confirming that smoking quantity matters more than binary smoking status.

**Panel 3 – Clinical Indicators**

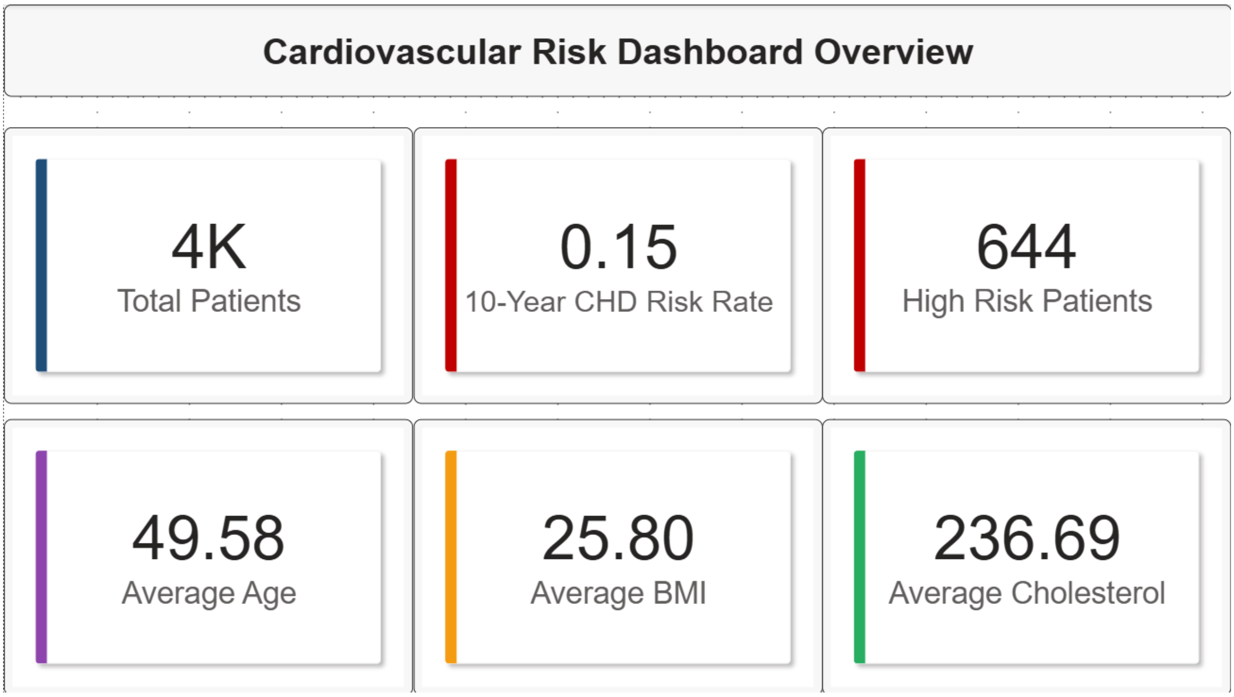
Visualizes sysBP and diaBP distribution by risk category (showing clear upward trend in high-risk patients), cholesterol threshold effect (spike at very high levels), and a diabetes vs non-diabetes CHD comparison. Note: the diabetes chart required careful aggregation to display risk rates rather than raw counts, to avoid misleading results from class size differences.

**Panel 4 – AI-Driven Analysis**

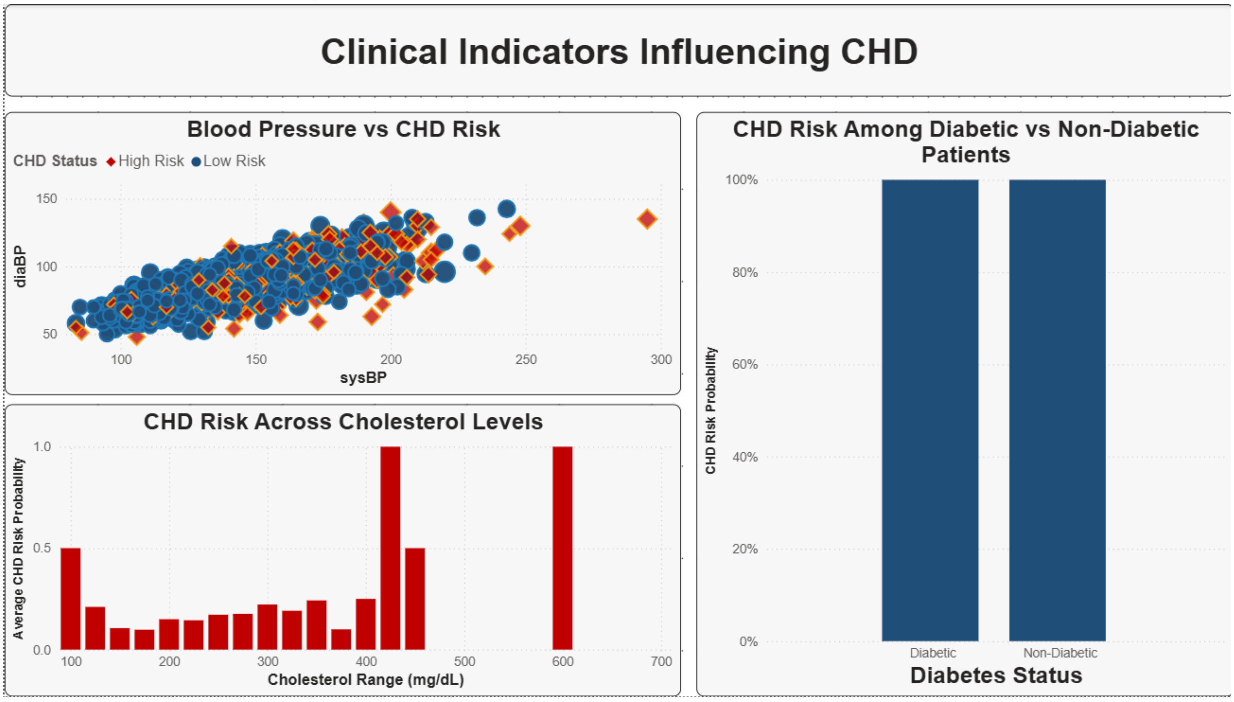
Incorporates Power BI's Key Influencers visual to identify the strongest threshold-based predictors, a top segments analysis segmenting the population into four risk tiers (Segment 1 carrying ~31.9% CHD rate, Segment 4 carrying ~9%), and a Decision Tree visualization showing the compounding multi-factor risk pathway: age > 60 → hypertension → heavy smoking → higher BMI → male → highest CHD risk.

### 7.4 Dashboard Screenshots

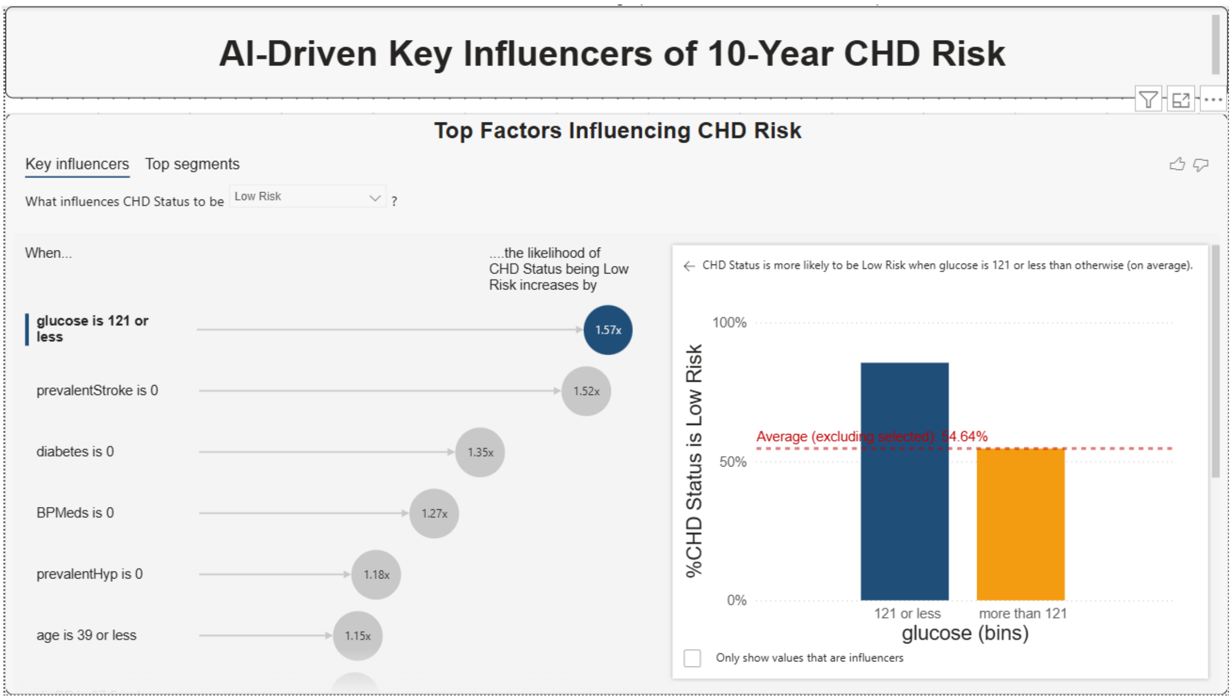
Please refer to Figures 20, 21, and 22 in the List of Figures. Dashboard screenshots are to be inserted at this location.




*[Figure 20: Power BI Dashboard – Overview Screenshot]*




*[Figure 21: Power BI Dashboard – Clinical Indicators Screenshot]*




*[Figure 22: Power BI Dashboard – AI-Driven Analysis Screenshot]*

# Chapter 8: Insights and Storytelling

## 8.1 Key Findings from Analysis

Across all analytical phases of this project, the following findings emerged with consistent evidence:

- Blood pressure is the dominant clinical predictor. Systolic blood pressure ranked #1 across all three feature selection methods (correlation, mutual information, random forest). Patients with sysBP above 176 mmHg face a CHD risk increase of approximately 2.88 times compared to those within normal ranges.
- Age acts as a risk gateway. Patients above age 64 show a risk multiplier of approximately 2.60 times. Beyond this threshold, the interaction between age and other risk factors (hypertension, smoking) creates a compounding effect — age does not raise risk linearly but accelerates it sharply in older cohorts.
- Glucose above 121 mg/dL is the single strongest nonlinear threshold predictor, elevating risk approximately 3.13 times. This reflects the metabolic damage of chronic hyperglycemia on arterial health.
- Diabetes and prevalent hypertension are the two binary features with the most dramatic impact. Diabetics face a CHD rate of ~37% versus ~15% for non-diabetics. Hypertensive patients face ~25% risk versus ~11%.
- Smoking quantity, not status, drives risk. Binary currentSmoker showed limited discriminative power, but cigsPerDay revealed a clear dose-response relationship. Heavy smokers (30–40 cigarettes/day) exhibit markedly elevated risk compared to light smokers.
- Cholesterol and BMI are secondary contributors. Their influence emerges primarily in tree-based and nonlinear models, suggesting interaction effects rather than direct linear causation.

## 8.2 Patterns and Trends Identified

The analysis revealed several structural patterns in the data:

- Multicollinearity cluster: sysBP, diaBP, and prevalentHyp form a tightly correlated group. PCA confirmed this by loading them together on PC1. In any predictive model, care must be taken to handle this redundancy (via regularization or using PCA components).
- Class imbalance pattern: The 85:15 split between non-CHD and CHD cases is consistent with real-world prevalence rates. This imbalance can bias classifiers toward predicting the majority class, making precision-recall metrics more informative than raw accuracy.
- Nonlinear risk thresholds: Multiple features showed threshold effects rather than linear gradients — glucose > 121, age > 64, sysBP > 176, diaBP > 110.5. These are better captured by tree-based models than linear regression.

### 8.3 Business and Social Insights

The findings carry meaningful implications for public health policy and clinical decision-making:

- Targeted screening programs: Resources should be concentrated on patients aged 60+, particularly those with hypertension or diabetes. Even a modest reduction in CHD rates within this high-priority cohort (Segments 1 and 2, representing ~60% of high-risk cases) would yield substantial public health benefit.
- Gender-aware interventions: Men are disproportionately represented in CHD positive cases (~19% male vs ~13% female CHD rate). Public health messaging and cardiovascular screening programs may need to be tailored differently by gender.
- Glucose monitoring as a prevention lever: The strong threshold effect at glucose > 121 mg/dL suggests that identifying and managing pre-diabetic patients before they cross this threshold could be a high-impact, cost-effective preventive strategy.
- Smoking cessation at scale: While binary smoking status has limited predictive power alone, the dose-response relationship confirms that smoking cessation programmes targeting heavy smokers would meaningfully reduce population-level CHD risk.

### 8.4 Decision-Making Based on Results

This analysis supports the following evidence-based decision pathways:

| Decision Context | Data-Driven Recommendation |
|---|---|
| Hospital triage prioritization | Flag patients aged 60+ with hypertension for mandatory cardiac screening |
| Resource allocation (clinics) | Invest in blood pressure monitoring and management infrastructure — highest ROI feature |
| Preventive pharmacology | Target patients with glucose > 121 and no diabetes diagnosis for glucose management |
| Population health programmes | Weight anti-smoking campaigns toward heavy smokers (30+ cigs/day) not just smokers in general |
| Insurance risk stratification | Age + sysBP + diabetes status forms a defensible, three-factor risk tier for underwriting |
| Follow-up scheduling | Prioritize Segments 1 & 2 from dashboard segmentation for quarterly follow-ups |

# Chapter 9: Conclusion

## 9.1 Summary of the Project

This project delivered a complete, end-to-end data science workflow applied to the Framingham Heart Study dataset, with the objective of predicting 10-year Coronary Heart Disease risk. Beginning with raw, uncleaned clinical data, the team systematically progressed through data loading, missing value imputation, outlier detection and capping, exploratory data analysis, feature engineering, advanced feature selection, dimensionality reduction via PCA, and interactive dashboard construction in Power BI.

## 9.2 Outcomes Achieved

- A fully cleaned dataset with zero missing values and controlled outliers was produced from a raw 4,240-record clinical dataset.
- Comprehensive EDA uncovered key risk associations — age, sysBP, diabetes, and prevalent hypertension are the strongest predictors of 10-year CHD risk.
- Feature engineering improved dataset quality through log1p transformation of cigsPerDay, ordinal encoding of education, and RobustScaler normalization.
- Three-method consensus feature selection reduced the feature set from 15 to 10 high-signal features with evidence-based justification.
- PCA confirmed feature redundancy, achieving ~80% variance retention in just 4 principal components. Component loading analysis revealed meaningful physiological groupings (cardiovascular pressure, metabolic, lipid-aging clusters).
- An interactive Power BI dashboard was built with four thematic panels: population KPIs, demographic/lifestyle risk, clinical indicators, and AI-driven analysis including Key Influencers, Top Segments, and Decision Tree visuals.
- Actionable clinical and public health insights were derived, including threshold-based risk markers and high-priority patient segments for intervention.

## 9.3 Limitations

- Class Imbalance: The 85:15 CHD-negative to CHD-positive ratio was not addressed through resampling (SMOTE, undersampling) in this project, which may limit the clinical utility of any predictive model trained on this data without further balancing.
- Dataset Age: The Framingham dataset reflects lifestyle and clinical patterns from decades ago. Contemporary risk factors (obesity rates, medication adherence, dietary shifts) may differ, reducing direct generalizability.
- No Predictive Model Deployed: This project stops at analysis and visualization. No classification model was trained, evaluated, or deployed. The feature selection and PCA work lays the groundwork but does not constitute a complete predictive system.
- Missing Value Volume in Glucose: Glucose had 388 missing values — the highest of any feature. Median imputation, while appropriate, introduces potential information loss given glucose's strong nonlinear predictive power.
- No External Validation: All findings are derived from a single dataset without cross-validation against an independent cohort, limiting the generalizability of insights.

### 9.4 Future Scope

- Model Development: Train and compare classification models (Logistic Regression, Random Forest, XGBoost, SVM) using the engineered feature set, with proper handling of class imbalance via SMOTE or class weighting.
- Risk Score Generation: Develop a composite risk score (akin to the Framingham Risk Score) using the top predictors identified in this project, enabling quantitative individual risk stratification.
- Streamlit Deployment: Build an interactive web application using Streamlit that accepts patient data as input and returns a predicted CHD risk score with feature-level explanation (SHAP values).
- Temporal Analysis: Investigate time-dependent risk progression by incorporating longitudinal follow-up data, enabling survival analysis (Kaplan-Meier, Cox regression).
- External Dataset Validation: Validate findings against the UCI Heart Disease dataset or other cardiovascular cohorts to test generalizability across populations.

# Chapter 10: References


## Dataset Sources

[1] Dileep, Heart Disease Prediction using Logistic Regression Dataset, Kaggle, 2019. Available: https://www.kaggle.com/datasets/dileep070/heart-disease-prediction-using-logistic-regression

[2] Amanajmera1, Framingham Heart Study Dataset, Kaggle. Available: https://www.kaggle.com/datasets/amanajmera1/framingham-heart-study-dataset

## Tools and Libraries

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[6] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," *Comput. Sci. Eng.*, vol. 9, no. 3, pp. 90–95, 2007.

[7] M. Waskom, "seaborn: statistical data visualization," *J. Open Source Softw.*, 6(60), 3021, 2021.

[8] Microsoft, *Power BI Documentation*. [Online]. Available: https://learn.microsoft.com/en-us/power-bi/

## Research Papers and Websites Referenced

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[11] GeeksforGeeks, Principal Component Analysis (PCA). Available: https://www.geeksforgeeks.org/data-analysis/principal-component-analysis-pca/

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# Appendix

## A. GitHub Repository

Project Repository:
https://github.com/Gokulnaath-gif/24ADI204_DSV_Team14
The repository contains all Python scripts (data cleaning, EDA, feature engineering, PCA), the Power BI dashboard file (.pbix), and the dataset used for analysis.

## B. Key Code Snippets

### B.1 Missing Value Imputation

```
# Separate numerical and categorical columns
num_cols = df.select_dtypes(include='number').columns.tolist()
cat_cols = df.select_dtypes(include='object').columns.tolist()
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
df[cat_cols] = df[cat_cols].fillna(df[cat_cols].mode().iloc[0])
```

### B.2 IQR Outlier Capping

```
def iqr_cap(df, col):
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    df[col] = df[col].clip(lower, upper)
    return df
```

### B.3 Log1p Transformation and RobustScaler

```
import numpy as np
from sklearn.preprocessing import RobustScaler
df['cigsPerDay'] = np.log1p(df['cigsPerDay'])
scaler = RobustScaler()
df[cont_cols] = scaler.fit_transform(df[cont_cols])
```

### B.4 PCA Application

```
from sklearn.decomposition import PCA
pca = PCA(n_components=0.95)  # Retain 95% variance
X_pca = pca.fit_transform(X_selected)
```